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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("K-Nearest Neighbours (K-NN)")
print("-----")
print("Training Samples :", len(X_train))
print("Testing Samples  :", len(X_test))
print("Accuracy          :", round(accuracy * 100, 2), "%")
sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = knn.predict(sample)
print("\nNew Sample Prediction:")
print("Predicted Class :", iris.target_names[prediction[0]])

```

K-Nearest Neighbours (K-NN)

Training Samples : 105

Testing Samples : 45

Accuracy : 100.0 %

New Sample Prediction:

Predicted Class : setosa
